

Shanitha Alochana

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PROFILE

A dedicated and responsible researcher committed for contributing to the field of wireless communication who possesses a set of strong work ethics, adaptability and a positive attitude with a background in Telecommunications, Cyber Security, Internet of Things, Machine Learning and Networking. A hardworking individual aspiring to leverage skills, knowledge and experience to contribute towards organizational objectives, demonstrating his own aptitude for rigorous academic inquiry.

WORK EXPERIENCE

Operations Support Systems Engineer

Subsidiary Elabram Telecommunications - Thailand

Colombo 05, Sri Lanka

June 2025–Present

- Operating the UME ElasticNet OSS platform for live network monitoring, control, and optimization.
- Supporting NGI (Network Geolocation Insight) analysis for event detection, coverage visualization, and identification of weak-signal areas.
- Preparing KPI and neighbor parameter reports to assist RF teams in troubleshooting and performance optimization.
- Monitoring alarms, counters, and KPIs to ensure network stability and high service quality.
- Executing antenna tilt and power tuning to balance coverage, enhance signal quality, and minimize interference.
- Delivering FAC (Final Acceptance Certificate) presentations and reports to stakeholders to support project acceptance and closure decisions.

RF Optimization Engineer

Subsidiary Elabram Telecommunications - Thailand

Colombo 05, Sri Lanka

Dec 2024–May 2025

- Performed SSV (Single Site Verification) activities for 2G, 3G, 4G, and 5G sites across multiple frequency bands.
- Verified site KPIs and parameter compliance with network acceptance criteria.
- Analyzed and compared radio parameters using drive test data to identify optimization opportunities.
- Optimized network coverage and signal quality by analyzing drive test results and implementing corrective actions.

Trainee International Voice and CEM DevOps Engineer

Dialog Axiata PLC, Sri Lanka

Colombo, Sri Lanka

Sep 2023–Apr 2024

- Gained hands-on experience with voice communication technologies, including VoIP and SIP.
- Strengthened troubleshooting and analytical skills through practical network and service performance monitoring.
- Performed trunk reconciliation analysis between Ribbon Softswitch and SBC billing systems.
- Supported CEM to GDE migration by coordinating access issue resolution with technical teams.
- Assisted in fine-tuning an IDD fraud detection tool by consolidating MSRN data from IR21 documents.
- Documented Huawei CEM metadata and data dictionaries to improve data governance.

EDUCATION

SLTC Research University

B.Sc. Eng. Hons. in Electronic and Telecommunication Engineering | **GPA - 2.91**

Meepe, Sri Lanka

Feb 2020 – Dec 2024

Bandarawela Central College

G.C.E Advanced Level Examination - Physical Science Stream

2-C passes and 1-S Pass

Bandarawela Central College

G.C.E Ordinary Level Examination

8-A passes and 1-B pass

Bandarawela, Sri Lanka

Mar 2017 – Aug 2019

Bandarawela, Sri Lanka

Jan 2011 – Dec 2016

SKILLS

Programming: Python, MATLAB, C, C++, Java, LaTeX

Hardware Technologies: Raspberry Pi, Arduino, NodeMCU (ESP8266), FPGA, LoRa, ESP32

Tools: VS Code, PyCharm, Microsoft Office, Node.js, WNG Analyzer, Elasticnet

Soft Skills: Problem-solving, Teamwork, Leadership, Adaptability, Fast learning ability, Collaboration, Flexibility

Languages: Sinhala, English

PROJECTS

- **Intelligent IoT-Based Forest Fire Detection and Early Warning System — Final Year Project**

Developed an intelligent IoT-based forest fire detection and early warning system using ESP32-based sensor nodes and LoRa SX1278 for long-range communication, with a Raspberry Pi used as the gateway for internet connectivity. Integrated temperature, humidity, CO, CO₂, and smoke sensors, implemented cloud data storage and visualization using ThingSpeak, applied machine learning for fire risk prediction, developed a real-time monitoring website, and enabled automated SMS alerts using the Vonage API.

- **Automated Water Motor System — Engineering Design Project, Aug 2022**

Developed an Arduino-based automated water motor control system to regulate water levels and reduce manual intervention using sensor-driven automation.

- **Beauty Services Web Application—Web-Based Application Development Project, Dec 2022**

Built a database-driven web application for salon appointment scheduling, service management and customer tracking using PHP, MySQL and HTML/CSS.

- **Hospital Management System— Software Engineering Project, Dec 2021**

Designed and implemented a software solution to manage hospital operations, including patient records, appointments and billing, improving workflow efficiency.

- **Remote Monitoring of Coke Cans in a Fridge with Real-time Updates on Smartphones, Apr 2022**

Built a real-time IoT-based fridge monitoring system to track temperature and beverage availability, providing 30-minute interval notifications to smartphones using MQTT.

INTERESTS

Telecommunication | Mathematics | Cyber Security | Internet of Things | Machine Learning | Networking | Programming | Data Science | Data Structures and Algorithms | Image Processing

PROFESSIONAL MEMBERSHIPS

Member of Electrical and Electronics Engineers (EEE)

Dec 2022 – Present

COURSES & CERTIFICATES

Cisco Certified Network Associate (CCNA) Certification

Offered by University of Moratuwa.

Oct 2023-Nov 2024

AWS Academy Graduate - AWS Academy Cloud Foundations

Offered by AWS Academy

Sep 2023-Dec 2023

REFERENCES

Mr.Premasiri Ratnayake
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Group Legal and Regulatory,
Dialog Axiata plc.
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